G.B. TECHNICAL UNIVERSITY, LUCKNOW
Study and Evaluation Scheme B. Tech in Electronics and Instrumentation, Instrumentation and Control, Applied Electronics and Instrumentation
[Effective from the session 2011-12]

YEAR 4th, SEMESTER-VII

S. No.	Course Code	SUBJECT	PERIODS			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM.			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1.	EOE-07*	Open Elective-I**	3	1	0	30	20	50	100	150	4
2.	EIC-02*	Departmental Elective-II	3	1	0	30	20	50	100	150	4
3.	EIC-701	Control System II	3	1	0	30	20	50	100	150	4
4.	EIC-702	Telemetry Principles	3	1	0	30	20	50	100	150	4
5.	EIC-703	Digital Measurement Techniques	3	1	0	30	20	50	100	150	4
6.	AUC-001	*Human Values & Professional Ethics	2	0	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
7.	EIC-751	Control System Lab II	0	0	2	-	20	20	30	50	1
8.	EIC-752	Telemetry Lab	0	0	3	-	50	50	-	50	2
9.	EIC-753	Industrial Training Viva-Voce	0	0	2	-	20	20	30	50	1
10.	EIC-754	Project	0	0	2	-	50	50	-	50	1
11.	GP-701	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**15**	**5**	**9**	**150**	**240**	**440**	**560**	**1000**	**26**

**** Open Electives-I**

1. EOE-071 Entrepreneurship Development
2. EOE-072 Quality Management
3. EOE-073 Operation Research
4. EOE-074 Introduction to Biotechnology

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YEAR 4th, SEMESTER-VIII

S. No.	Course Code	SUBJECT	PERIODS			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM.			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1.	EOE-08*	Open Elective-II**	3	1	0	30	20	50	100	150	4
2.	EIC-03*	Departmental Elective-III	3	1	0	30	20	50	100	150	4
3.	EIC-801	Optimal Control Systems	3	1	0	30	20	50	100	150	4
4.	EIC-802	Biomedical Instrumentation	3	1	0	30	20	50	100	150	4
5.	AUC-001	*Human Values & Professional Ethics	2	0	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
6.	EIC-851	Project	0	0	12	-	100	100	250	350	8
7.	GP-801	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**12**	**4**	**12**	**120**	**180**	**350**	**650**	**1000**	**24**

**** Open Electives-II**

1. EOE-081 Non Conventional Energy Resources
2. EOE-082 Nonlinear Dynamic Systems
3. EOE-083 Product Development
4. EOE-084 Automation & Robotics

LIST OF ELECTIVES:

Elective – II

1. EIC-021 Optical Instrumentation
2. EIC-022 Power Plant Instrumentation
3. EEC-023 ANN
4. EEC-024 Filter Design

Elective – III

1. EIC-031 Computerized Process Control
2. EIC-032 Bio-medical Signal Processing
3. EIC-033 Analytical Instrumentation
4. EIC-034 Micro and Smart Systems
5. EEC-032 Digital System Design using VHDL

EIC–701 CONTROL SYSTEMS II			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	Sampling and Signal Conversion: Sampled-Data Control Systems, Digital to Analog Conversion, Sample and Hold operations, Sample and Hold Devices, frequency–Domain Characteristic of Zero order Hold. **The Z-Transform:** Linear Difference equations, The Pulse Response, The Definition of the Z-transform, Relationship between the Laplace transform and the Z-transform, Relationship between S-plane and the Z-plane, The constant-Damping Loci, The constant- Frequency Loci, The constant-Damping Ratio Loci, The Inverse Z-Transform, Theorems of the Z-transform, Limitations of the Z-transform, Application of the Z-transform ,Stability Analysis, Systems with Dead-Time.	1/1, 2/2 1/2, 2/3	4 6
II	Transfer Functions, Block Diagrams, and Signal flow Graphs The Pulse Transfer Function and The Z-Transfer Function, The Pulse Transfer Function of the Zero-Order Hold and the Relation Between G(s) and G(z), Closed loop systems, The Sampled Signal flow Graph, The Modified Z-transfer function, Multirate Discrete Data System. Transform Design of Digital Controls Design of position Servo Design Specifications, Design on the W- plane, Design of the W-plane, the Digital PID Controllers.	2/4 1/3	6 4
III	State Space Analysis of Sampled Data Systems Discrete time state equations. Similarity Transformations, The Cayley-Hamilton Theorem, Realization of Pulse Transfer function, State Equations for sampled Data Systems, Concepts of Controllability and Observability, Liapunov Stability Analysis Systems with Dead time.	1/4, 2/6	7
IV	Design of digital controls using State Space analysis Formulation of the optimal control Problem Optimal State Regulator, Use of State Regulator results, Eigen value Assignment by State feedback, State observers Stochastic optimal State Estimation.	1/5	6
V	Mechanization of Control algorithms Using Micro Processors General Description of Microcontrollers, Digital quantization, Microprocessor based Position Control System.	1/7	7

Text Books:

1. M. Gopal, “Digital Control Engineering”, New Age International Publishers.
2. B.C. Kuo ,“Digital Control Systems”, Oxford University Press.

EIC-702 Telemetry Systems			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	Introduction to Telemetry Principles: Basic System, Classification, Non electrical telemetry systems, Voltage and current Telemetry systems, Frequency Telemetering, Power line carrier Communication.	1/1	4
II	Multiplexed System: Frequency Division Multiplex System- FDM, IRIG Standards, FM circuits, Phase Modulation Circuits, Receiving end, Phase Locked Local Loop, Mixers. Time Division Multiplexed System – TDM/PAM system, PAM/ PM systems, TDM- PCM System, Digital Multiplexer, PCM Reception, Coding for varying level, DPCM, Standards.	1/4,1/5	10
III	Modems: Modems Introduction, QAM, modem protocol.	1/6	4
IV	Transmitter and Receiver: Transmitters, Transmission Techniques, Inter stage Coupling, Receiver Antennas: The Ideal structure, dipoles, arrays, current distribution and design consideration, Microwave Antennas.	1/7, 1/9	10
V	Filters: Polynomial, Filters, Active RC Filters, Universal Filter Circuits, Switched Capacitor Filters, Digital Filters Basics of Satellite and Fiber Optic Telemetry Data Acquisition Systems (DAS), µP based DAS, Remote Control	1/11 1/12, 1/13 1/14	2 8 2

Text Books:

1. D Patranabis, Telemetry Principle; TMH Ed 1 1999.

EIC-703 Digital Measurement Techniques			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	Philosophy of digital measurements. **Digital Time Measurement Techniques:** Measurement of time interval between two events, error in time interval measurement, Vernier technique for small time measurement, measurement of time interval with constraints, measurement of periodic time, phase, time interval between two events defined by voltage levels, capacitance, quality factor of ringing circuit, decibel meter, logarithmic A/D converter.	1	8
II	**Digital Frequency Measurement Techniques:** Measurement of frequency, ratio of two frequencies, product of two frequencies, high frequency, average frequency difference, deviation of power frequency, peak frequency. Fast low-frequency measurement.	2	8
III	Digitally Programmable Circuits: Resistor, Potentiometer, amplifiers, Schmitt trigger, dual polarity gain amplifiers. Programmable gain amplifier with dual output, two stage programming, programmable biquads.	3	8
IV	Digital to Analog Converters: Output input relation, DACs derived from programmable gain amplifiers, Weighted-resistor DAC, Weighted current DAC, Weighted reference voltage DAC, Ladder DAC, switches.	4	8
V	**Digital Voltage Measurement Techniques:** Sampling theorem, time-division multiplexing, quantization, indirect type A/D converters, direct type A/D converters, Input circuitry of a digital voltmeter.	5	8

Text Book:

1. T. S. Rathore, “Digital Measurement Technique”, Narosa Publishing House, 1996.

ELECTIVES II

EIC-021 OPTICAL INSTRUMENTATION		3 1 0
UNIT	TOPICS	LECTURES
I	Light Sourcing, Transmitting and Receiving Concept of Light, Classification of different phenomenon based on theories of light, Basic light sources and its Characterization, Polarization , Coherent and Incoherent sources, Grating theory ,Application of diffraction grating, Electro-optic effect, Acousto-optic effect and Magneto-optic effect.	8
II	Opto –Electronic devices and Optical Components Photo diode, PIN, Photo-Conductors, Solar cells, ,Phototransistors, Materials used to fabricate LEDs and Lasers Design of LED for Optical communication, Response times of LEDs ,LED drive circuitry, Lasers Classification :Ruby lasers, Neodymium Lasers, He- Ne Lasers, CO_2 Lasers, Dye Lasers, Semiconductors Lasers, Lasers Applications.	8
III	Interferometry Interference effect, Radio-metry, types of interference phenomenon and its Application, Michelson's Interferometer and its application Fabry-perot interferometer, Refractometer, Rayleigh's interferometers, Spectrographs and Monochromators, Spectrophotometers, Calorimeters, Medical Optical Instruments	8
IV	Holography: Principle of Holography, On-axis and Off axis Holography, Application of Holography, Optical data storage. Optical Fiber Sensors: Active and passive optical fiber sensor, Intensity modulated, displacement type sensors, Multimode active optical fiber sensor (Micro bend sensor)Single Mode fiber sensor-Phase Modulates and polarization sensors	8
V	Fiber optic fundamentals and Measurements: Fundamental of Fibers, Fiber Optic Communication system, Optical Time domain Reflectometer (OTDR), Time domain dispersion measurement, Frequency Domain dispersion measurement, Laser Doppler velocimeter,	8

Text Book:

1. J. Wilson & J. F. B. Hawkes, "Optoelectronics: An Introduction" PHI/ Pearson
2. Rajpal S. Sirohi "Wave Optics and its Application", Hyderabad, Orient longman Ltd.
3. A. Yariv, "Optical Electronics", C. B. S. Collage Publishing, New York, 1985.

EIC-022 POWER PLANT INSTRUMENTATION		3 1 0
UNIT	TOPICS	LECTURES
I	Energy sources, their availability, worldwide energy production, energy scenario of India. Introduction to Power generation- Classification: Renewable and non-renewable energy generation resources. Renewable: small hydro; modern biomass; wind power; solar; geothermal and bio-fuels. Non renewable: fossil fuels (coal, oil and natural gas) and nuclear power. Boiler: Types of boilers, boiler safety standards. Boiler instrumentation, control and optimization, combustion control, air to fuel ratio control, three element drum level control, steam temperature and pressure control, boiler interlocks, sequence event recorder, data acquisition systems.	8
II	Thermal Power Plant- Method of power generation, layout and energy conversion process, Types of Turbines & control, Types of Generators, condensers. Types of pumps and Fans, variable speed pumps and Fans, Material handling system, study of all loops-water, steam, fuel etc.	8
III	Hydroelectric Power Plant- Site selection, Hydrology, Estimation electric power to be developed, classification of Hydropower plants, Types of Turbines for hydroelectric power plant, pumped storage plants, storage reservoir plants.	8
IV	Wind Energy: Power in wind, Conversion of wind power, Aerodynamics of wind turbine, types of wind turbine, and modes of operation, power control of wind turbines, Betz limit, Pitch & Yaw control, wind mill, wind pumps, wind farms, different generator protections, data recording, trend analysis, troubleshooting & safety. Solar Energy: solar resource, solar energy conversion systems: Solar PV technology: Block diagram of PV system, advantages and limitations. Solar thermal energy system: Principle, solar collector and its types, solar concentrator and its types, safety.	8
V	Nuclear Power Plant: Nuclear power generation, control station and reactor control. **Comparison of various plants:** Comparison of thermal power plant, hydro electric power plant, wind, solar, nuclear power plant on the basis of: Performance, efficiency, site selection, Economics-capital and running, safety standards, pollution, effluent management and handling. Power plant safety, Pollution monitoring, control Sound, Air, smoke, dust, study of Electrostatic precipitator.	8

Text Books:

1. G.F. Gilman, “Boiler Control Systems Engineering”, ISA Publication.
2. P. K. Nag, “Power plant engineering”, McGraw Hill.

Reference Books:

1. B. H. Khan, “Non-conventional energy resources”, McGraw Hill.
2. Chetan Singh Solanki, “Renewable energy Technology”, Prentice Hall Publication.
3. S. P. Sukhatme, “Solar Energy”, Tata McGraw Hill.
4. G. D. Rai, “Nonconventional energy sources”, Khanna Publication.

EEC-023 Artificial Neural Networks		3 1 0
UNIT	TOPICS	LECTURES
I	Introduction: Introduction and history, human brain, biological neuron, models of neuron, signal flow graph of neuron, feedback, network architecture, knowledge representation, Artificial intelligence and neural networks. Learning Process: Error correction learning, memory based learning, Hebbian learning, competitive learning, Boltzmann learning, learning with and without teacher, learning tasks, memory and adaptation.	4 4
II & III	Artificial neurons, Neural networks and architectures, introduction, neuron signal function, mathematical preliminaries, Feed forward & feedback architecture. Geometry of Binary threshold neurons and their networks, Pattern recognition, convex sets and convex hulls, space of Boolean functions, binary neurons for pattern classification, non linear separable problems, capacity of TLN, XOR solution. Perceptrons and LMS Learning objective of TLN, pattern space & weight space, perceptron learning algorithm, perceptron convergence theorem, pocket algorithm, α – LMS learning, MSE error surface, steepest descent search, μ – LMS and application. Back propagation and other learning algorithms Multilayered architecture, back propagation learning algorithm, practical considerations, structure growing algorithms, applications of feed forward neural networks, reinforcement learning.	2 3 5
IV	Statistical Pattern Recognition Bayes' theorem, classical decisions with Bayes' theorem, probabilistic interpretation of neuron function, interpreting neuron signals as probabilities, multilayered networks & posterior probabilities, error functions for classification problems. RBF Networks Regularization networks, generalized RBF networks, RBF network for solving XOR problem, comparison of RBF networks & multilayer perceptrons. Stochastic Machines: Statistical mechanics, simulated annealing, Boltzmann machine.	4 2
V	Adaptive Resonance Theory Building blocks of adaptive resonance, Adaptive Resonance Theory 1. Self Organizing Feature MAP Introduction, Maximal eigenvector filtering, principal component analysis, generalized learning laws, competitive learning, vector quantization, Mexican hat networks.	8

Text Books:

1. Kumar Satish, "Neural Networks", TMH.
2. Simon Haykin, "Neural Networks", PHI.

EEC-024 Filter Design		3 1 0
UNIT	TOPICS	LECTURES
I	Review of op-amps circuits, Categorization of filters-Low-pass filter, High-pass filter, band-pass filter, band-reject filter, Gain equalizers, and Delay equalizers.	8
II	Approximation Theory: Butterworth approximation, Chebyshev approximation, Inverse Chebyshev approximation, Basic of sensitivity, Frequency Transformations.	8
III	Three amplifier Biquad: Basic low pass and band pass circuit, realization of the general Biquadratic Functions, summing of four Amplifier biquad, feed forward three amplifier biquad, Passive Ladder structures, Inductor Substitution using Gyrator, Transformation of elements using the FDNR. Active ladder filters. Active R filters.	10
IV	Elementary transconductor building blocks, resistors, integrators, amplifiers, summers, gyrator, First and second order filters, higher order filters.	8
V	Switched capacitor filters: The MOS switch, The switched capacitor, first order building blocks, second order sections, sampled data operation, Switched capacitor first and second order filters, Bilinear transformation based SC filter design.	6

Text Books:

1. Gobind Daryanani, “Principles of active network synthesis and design”,John Wiley & Sons.
2. R. Schaumann, M. E. Van Valkenburg, “Design of analog filters”, Oxford University Press.

EIC 751 DIGITAL CONTROL SYSTEM LAB II

Minimum of 10 experiments to be performed

At least four experiments to be done on MATLAB

Suggestions are -

1. Discrete Time LTI model
2. Discrete pole locations & transients response
   Small damping ($\varepsilon = 0.1$ $W_n = 4\pi/5T$) Medium damping ($\varepsilon = 0.4$ $W_n = 11\pi/5T$) Large damping ($\varepsilon = 0.8$ $W_n = \pi/4T$)
3. Digital DC motor Speed control with PID controller
4. Designing Lead & Lag Compensators
5. Kalman Filter design
6. State space design for the Inverted pendulum

EIC 752 TELEMETRY LAB

Minimum of 10 experiments to be performed

1. Measurement of Temperature Using RTD/ Thermister and amplification to an appropriate level suitable for Tele transmission.
2. Sampling through a S/H Circuit and reconstruction of the sampled signal. Observe the effect of sampling rate & the width of the sampling pulses.
3. Realization of PCM signal using ADC and reconstruction using DAC using 4-bit/8 bit systems. Observe the Quantization noise in each case.
4. Fabricate and test a PRBS Generator.
5. Realization of data in different formats such as NRZ-L, NRZ-M and NRZ-S.
6. Clock recovery circuit from NRZ-L data using PLL.
7. Manchester coding & decoding (Biphase L) of NRZ-L Data.
8. Coding and decoding NRZ-L into URL-L (Unipolar return to Zero coding).
9. ASK – Modulation and Detection
10. FSK – Modulation and Detection
11. PSK - Modulation and Detection.
12. Error introduction, Error Detection & Correction using Hamming Code.
13. Amplitude modulation and Detection of signal obtained from experiment no.1.

THEORY CORE 8th SEMESTER

EIC–801 Optimal Control			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	General Mathematical Procedures: Formulation of the optimal control Problem, Calculus of variations, Minimum principle, Dynamic Programming, Numerical Solution of Two-point Boundary value problem.	1/10, 2/11	8
II	Optimal Feedback Control: Discrete-Time linear State regulator, Continuous-Time Linear state Regulator results of solve other linear problems, Suboptimal Linear regulators, Minimum-time Control of Linear Time-Invariant System.	1/11	8
III	Stochastic Optimal Linear Estimation and Control Stochastic processes and linear systems, Optimal Estimation for Linear Discrete time Systems Stochastic Optimal Linear Regulator,	1/12	8
IV & V	Microprocessor and DSP control Basic computer Architecture, Microprocessor Control of Control System, Single Board Controllers with Custom Designed Chips, Digital Signal Processors, Effect of finite World Length and Quantization on Controllability and Closed Loop –Pole Placement, Effects of Quantization, and Time Delays in Microprocessor Based control systems.	2/12	16

Text Books:

1. M. Gopal, “Modern Control Engineering”, New Age International Publishers.
2. B.C. Kuo, “Digital Control Systems”, Oxford University Press

Reference Book:

1. Brain D.O. Anderson, John B. Moore, “Optimal control Linear Quadratic Methods”, Prentice Hall of India Private Limited

EIC-802 BIO-MEDICAL INSTRUMENTATION			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	Introduction: Specifications of bio-medical instrumentation system, Man-Instrumentation system Components, Problems encountered in measuring a living system. Basics of Anatomy and Physiology of the body.	1/1 , 2/1 &2	3
	Bioelectric potentials: Resting and action potentials, propagation of action potential, The Physiological potentials – ECG, EEG, EMG, ERG, EOG and Evoked responses.	1/2 , 2/3	3
	Electrodes and Transducers: Electrode theory, Biopotential Electrodes – Surface electrodes, Needle electrodes, Microelectrodes, Biomedical Transducer.	2/4 & 16, 3/4	2
II	Cardiovascular Measurements: Electrocardiography – ECG amplifiers, Electrodes and Leads, ECG –Single channel, Three channel, Vector Cardiographs, ECG System for Stresses testing, Holter recording, Blood pressure measurement, Heart sound measurement. Pacemakers and Defibrillators.	2/7 & 10 , 3/5 & 6	6
	Patient Care & Monitoring: Elements of intensive care monitoring, displays, diagnosis, Calibration & Reparability of patient monitoring equipment.	1/ 6	2
III	Respiratory system Measurements: Physiology of Respiratory system. Measurement of breathing mechanism – Spirometer. Respiratory Therapy equipments: Inhalators, Ventilators & Respirators, Humidifiers, and Nebulizers & Aspirators.	2/8, 3/8	5
	Nervous System Measurements: Physiology of nervous system, Neuronal communication, Neuronal firing measurements.	3/10	3
IV	Ophthalmology Instruments: Electroretinogram, Electro - oculogram, Ophthalmoscope, Tonometer for eye pressure measurement.	2/7 & 13	2
	Diagnostic techniques: Ultrasonic diagnosis, Eco - cardiography, Eco-encephalography, Ophthalmic scans, X-ray & Radio-isotope diagnosis and therapy, CAT-Scan, Emission computerized tomography, MRI.	1/19, 22 & 23, 2/12 &14	6
V	Bio-telemetry: The components of a Bio-telemetry system, Implantable units, Telemetry for ECG measurements during exercise, for Emergency patient monitoring.	1/9, 3/12	4
	Prosthetic Devices and Therapies: Hearing Aides, Myoelectric Arm, Dia-thermy, Laser applications in medicine.	1/17, 2/7, 10 &13	4

TEXT BOOKS:

1. R. S. Khandpur, "Biomedical Instrumentation", TMH
2. S. K. Venkata Ram, "Bio-Medical Electronics &Instrumentation (Revised)", Galgotia.
3. J. G. Webster (editor), "Medical Instrumentation Application & Design", 3rd Ed WILEY, India

REFERENCE BOOKS:

1. Cromwell, "Biomedical Instrumentation and Measurements" PHI
2. J. G. Webster, "Bio- Instrumentation", Wiley
3. S. Ananthi, "A Text Book of Medical Instruments", New Age International
4. Carr & Brown, "Introduction to Biomedical Equipment Technology", Pearson
5. Pandey & Kumar, "Biomedical Electronics and Instrumentation", Kataria

ELECTIVE

EIC-031 COMPUTERISED PROCESS CONTROL			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	Basics of Computer-Aided Process Control: Role of computers in process control, Elements of a computer aided Process control System, Classification of a Computer –Aided Process Control System Computer-Aided Process–control Architecture: Centralized Control Systems, Distributed control Systems, Hierarchical Computer control Systems. Economics of Computer-Aided Process control. Benefits of using Computers in a Process control. Process related Interfaces: Analog Interfaces, Digital Interfaces, Pulse Interfaces, Standard Interfaces	1/4	8
II	Industrial communication System: Communication Networking, Industrial communication Systems, Data Transfer Techniques, Computer Aided Process control software, Types of Computer control Process Software, Real Time Operating System.	1/5	8
III	Process Modelling for computerized Process control: Process model, Physical model, Control Model, Process modeling. Modelling Procedure: Goals Definition, Information Preparation, Model Formulation, Solution Finding, Results Analysis, Model Validation	1/2	8
IV	Advanced Strategies For Computerised Process control: Cascade Control, Predictive control, Adaptive Control, Inferential control, Intelligent Control, Statistical control.	1/7	8
V	Examples of Computerized Process Control: Electric Oven Temperature Control, Reheat Furnace Temperature control, Thickness and Flatness control System for metal Rolling, Computer-Aided control of Electric Power Generation Plant.	1/10	8

Text Book:

1. S. K. Singh, “Computer Aided Process control”, PHI.

Reference Books:

2. C. L. Smith, “Digital computer Process Control”, Ident Educational Publishers.
3. C. D. Johnson, “Process Control Instrumentation Technology”, PHI.
4. Krishan Kant, “Computer Based Industrial Control”
5. Pradeep B. Deshpande & Raymond H. Ash, “Element of Computer Process Control with Advance Control Applications”, Instrument Society of America, 1981.
6. C. M. Houpis & G. B. Lamond, “Digital Control System Theory”, Tata McGraw Hill.

EIC-032 BIOMEDICAL SIGNAL PROCESSING			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	Introduction to Bio-Medical Signals: Classification, Acquisition and Difficulties during Acquisition. Basics of Electrocardiography, Electroencephalography, Electromyography & electro-retinography Role of Computers in the Analysis, Processing, Monitoring & Control and image reconstruction in bio-medical field.	3/1& 3 1/2, 3/3 1/1, 3/15	2 4 2
II	ECG: Measurement of Amplitude and Time Intervals, QRS Detection (Different Methods), ST Segment Analysis, Removal of Baseline Wander And Power line Interferences, Arrhythmia Analysis, Portable Arrhythmia Monitors.	1/12 & 13, 2/ 7	8
III	Data Reduction: Turning Point algorithm, AZTEC Algorithm, Fan Algorithm, Huffman and Modified Huffman Coding, Run Length Coding.	1/10	8
IV	EEG: Neurological Signal Processing, EEG characteristic, linear prediction theory, Sleep EEG, Dynamics of Sleep/Wake transition. Study of pattern of brain waves, Epilepsy-Transition, detection and Estimation. EEG Analysis By Spectral Estimation: The Bt Method, Periodogram, Maximum Entropy Method & AR Method, Moving Average Method. The ARMA Methods, Maximum Likelihood Method.	2/5 2/4	2 6
V	EP Estimation: by Signal Averaging, Adaptive Filtering:- General Structures of Adaptive filters, LMS Adaptive Filter, Adaptive Noise Cancelling, Wavelet Detection:- Introduction, Detection By Structural features, Matched Filtering, Adaptive Wavelet Detection, Detection of Overlapping Wavelets.	1/8 & 9 2/6 & Appendix B	8

Text Books:

1. Willis J. Tomkin, “Biomedical Digital Signal Processing”, PHI.
2. D. C. Reddy, “Biomedical Signal Processing”, McGraw Hill
3. Crommwell,Weibel and Pfeifer, “Biomedical Instrumentation and Measurement”, PHI

Reference Books:

1. Arnon Cohen, “Biomedical Signal Processing (volume-I)”, Licrc Press
2. Rangaraj M. Rangayyan, “Biomedical Signal Analysis A Case Study Approach”, John Wiley and Sons Inc.
3. John G. Webster, “Medical instrumentation Application and Design”, John Wiley & Sons Inc.

EIC-033 Analytical Instruments			3 1 0
UNIT	TOPICS	Text No./ Chapter	LECTURES
I	UV – Visible Spectroscopy: Introduction, Electromagnetic Radiation, Laws relating to absorption radiation, Absorption Instruments, Ultraviolet and visible absorption spectroscopy, Calorimeters, Double Beam spectrophotometer (Optical Diagram & Block Diagram) Microprocessor based Spectrophotometer (Block Diagram)	1/2	8
II	Infrared Spectroscopy, Basic Components of IR Spectrophotometers, Type of Infrared Spectrophotometers, Sample Handling Techniques.	1/3	4
III	Flame photometers: principle, constructional details of flame photometers, types of flame photometers, types of flame photometers, clinical flame photometers, accessories for flame photometer, expression for concentration, interferences in flame photometry, procedure for determinations. Atomic Absorption Spectrometers: Atomic Absorption Spectroscopy, Atomic Absorption Instrumentation, Sources of interferences, meter scale.	1/4 1/5	4 4
IV	Mass Spectrometers: Basic Mass Spectrometer, Principle of operation, Type of Mass Spectrometers, components of Mass Spectrometers, inductively coupled plasma-mass spectrometer, trapped ion analyzers, ion cyclotron resonance (ICR) mass spectrometer, quadruple ion trap mass spectrometer, applications of mass spectrometry, gas chromatograph-mass spectrometer, liquid chromatograph-mass spectrometer, tandem mass spectrometry (MS/MS).	1/9	8
V	Nuclear Magnetic Resonance (NMR) Spectroscopy, Principle of NMR, types of NMR spectrometers, constructional details of NMR spectrometer, variation T-60A NMR spectrometer, sensitivity enhancement for analytical NMR-spectroscopy, Fourier transform NMR spectroscopy.	1/10	8

Text Books:

1. R. S. Kandpur, “Handbook Of Analytical Instruments”, TMH 2nd Edition,
2. Willard, Merritt, Dean and Settle, “Instrumental Methods of Analysis”, 7th Edition, CBS Publishers.

EIC-034 MICRO AND SMART SYSTEMS		3 1 0
UNIT	TOPICS	LECTURES
I	Introduction, Why miniaturization?, Microsystems versus MEMS, Why micro fabrication?, smart materials, structures and systems, integrated Microsystems, applications of smart materials and Microsystems,.	5
II	Micro sensors, actuators, systems and smart materials: Silicon capacitive accelerometer, piezoresistive pressure sensor, conductometric gas sensor, an electrostatic combo-drive, a magnetic microrelay, portable blood analyzer, piezoelectric inkjet print head, micromirror array for video projection, smart materials and systems.	8
III	Micromachining technologies: silicon as a material for micro machining, thin film deposition, lithography, etching, silicon micromachining, specialized materials for Microsystems, advanced processes for micro fabrication.	8
IV	Modeling of solids in Microsystems: Bar, beam, energy methods for elastic bodies, heterogeneous layered beams, bimorph effect, residual stress and stress gradients, poisson effect and the anticlastic curvature of beams, torsion of beams and shear stresses, dealing with large displacements, In-plane stresses. Modelling of coupled electromechanical systems: electrostatics, Coupled Electro-mechanics: statics, stability and pull-in phenomenon, dynamics. Squeezed film effects in electro-mechanics.	8
V	Integration of micro and smart systems: integration of Microsystems and microelectronics, microsystems packaging, case studies of integrated Microsystems, case study of a smart-structure in vibration control. Scaling effects in Microsystems: scaling in: mechanical domain, electrostatic domain, magnetic domain, diffusion, effects in the optical domain, biochemical phenomena.	

Text book:

1. G. K. Ananthasuresh, K. J. Vinoy, S. Gopalakrishnan, K. N. Bhat and V. K. Atre, “Micro and smart systems”, Wiley India, 2010.

EEC-032 DIGITAL SYSTEM DESIGN USING VHDL		3 1 0
Unit	Topic	Lectures
I	Introduction to VHDL, reserve words, structures, modeling, objects, data type and operators, sequential statements and processes, sequential modeling and attributes, conditional assignment, concatenation and case, array loops and assert statements, subprograms.	8
II	Digital System Design Automation– Abstraction Levels, System level design flow, RTL design flow, VHDL. RTL Design with VHDL – Basic structures of VHDL, Combinational circuits, Sequential circuits, Writing Test benches, Synthesis issues, VHDL Essential Terminologies VHDL Constructs for Structures and Hierarchy Descriptions – Basic Components, Component Instantiations, Iterative networks, Binding Alternatives, Association methods, generic Parameters, Design Configuration	8
III	Concurrent Constructs for RT level Descriptions – Concurrent Signal Assignments, Guarded signal assignment Sequential Constructs for RT level Descriptions – Process Statement, Sequential WAIT statement, VHDL Subprograms, VHDL library Structure, Packaging Utilities and Components, Sequential Statements. VHDL language Utilities - Type Declarations and Usage, VHDL Operators, Operator and Subprogram overloading, Other TYPES and TYPE – related issues, Predefined Attributes	8
IV	VHDL Signal Model – Characterizing hardware languages, Signal Assignments, Concurrent and Sequential Assignments, Multiple Concurrent Drivers Standard Resolution	8
V	Hardware Cores and Models - Synthesis rules and styles, Memory and Queue Structures, Arithmetic Cores, Components with Separate Control and Data parts. Core Design Test and Testability - Issues Related to Design Test, Simple Test benches.	8

TEXT BOOKS:

1. Z. Navabi, “VHDL-Modular Design and Synthesis of cores and Systems”, TMH – 3rd Edition.
2. R.D.M. Hunter, T. T. Johnson, “Introduction to VHDL” Spriger Publication, 2010.

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1. J. Bhaskar, “A VHDL Primer”, Addison Wesley, 1999.
2. M. Ercegovac, T. Lang and L.J. Moreno, ”Introduction to Digital Systems”, Wiley,2000
3. C. H. Roth, “Digital System Design using VHDL”, PWS Publishing
4. Enoch O. Hwang , “Digital Logic and Microprocessor design with VHDL” – Thomson publication.